ORCA

Marine Ecosystem Reasoning with Collaborative Agents

Team briefing document — SIH26176 | ISRO | Space Technology

1. Problem statement

Official reference: SIH26176, submitted by ISRO, theme Space Technology / Marine Ecosystem.

India's marine ecosystem supports millions of livelihoods across fishing, coastal communities, and shipping, yet the satellite and oceanographic data that could protect and guide them, chlorophyll levels, sea temperature, tides, weather hazards, sits scattered across specialist portals in technical formats disconnected from each other. There is no accessible, intelligent system that reasons across these sources and responds in plain, trustworthy language, in the user's own language.

2. What we are building, in one paragraph

An agentic AI-powered conversational platform for marine intelligence. Users ask natural-language questions, such as whether it is safe to fish near a location tomorrow, and a team of specialised AI agents pulls real satellite and oceanographic data, reasons across it, and answers in plain language, in the user's own language, alongside an interactive map. The same engine serves fishermen, coastal patrol officers, maritime operators, coastal authorities, and researchers.

3. Novelty — what makes this different

- Collaborative multi-agent reasoning, not a single model or static dashboard.
- Conversational-first design, the map is a supporting visualisation the agent produces, not the primary interface on its own.
- Automatic multilingual reasoning, detects the query's language and responds natively, no manual language picker.
- Explainable verdicts with a confidence score and a cited evidence trail, never a black-box answer.
- One engine, five stakeholder groups, reframed per audience rather than five separate tools.
- Built on India's own ISRO and INCOIS data infrastructure.

4. Core features and outputs

Per location or region, the system produces:

- Chlorophyll level (mg/m3) and sea surface temperature, current reading
- Algae bloom risk verdict: low, medium, or high, compared to that location's own historical baseline, not a global threshold
- Fishing favourability verdict: favourable, caution, or avoid
- A 7 to 14 day historical trend chart
- A 2 to 3 day forecast with a confidence band that widens further into the future
- A confidence score (0 to 100) and a plain-language evidence trail explaining the verdict
- Tide, wave, and wind conditions for that location

- Short-range route safety check between two points, up to roughly 100 to 150 km, 1 to 2 day window
- Geofencing alerts near India's EEZ boundary and marine protected areas
- Proactive hazard alerts for cyclones, lightning, and high waves
- A formatted, advisory-style output, similar to official INCOIS or IMD bulletins

5. Interface design

Two ways to query one shared engine, side by side in a single screen.

- Manual mode: user zooms, pans, or clicks any point or region directly on the map, and the data panel opens immediately.
- Chat mode: user types or speaks a question. The Planning Agent resolves the location and intent, other agents fetch and analyse data, and the Reporting Agent replies in the user's language. The map automatically pans to and highlights the relevant zone, using the exact same underlying function as a manual click.
- Visual reference: Zoom Earth (zoom.earth/maps/temperature) for the interactive map feel, colour gradient overlay, and clean legend style we are aiming for.
- Map library: Leaflet, not Google Maps, chosen for the clean scientific look, no API key dependency, and because it is the standard choice for oceanographic tools of this kind.

6. Data sources — verified

Variable	Source	Access status
Chlorophyll	MODIS / VIIRS via INCOIS LAS (OPeNDAP)	Confirmed, real-time, no restriction
Sea surface temperature	INCOIS Ground Station (AVHRR)	Confirmed, verified with real 2026 sample files
Bloom indices	INCOIS pre-computed bloom indices	Confirmed, since March 2010
Waves	Wave Watch III via INCOIS	Confirmed, near real-time
Currents, wind	INCOIS (GODAS-MOM, OSCAT, drifting buoys)	Confirmed
Tides	TideCheck API (516 Indian stations)	Confirmed free fallback; INCOIS tide gauge access still unverified
EEZ boundary	marineregions.org	Confirmed, free GeoJSON
Marine protected areas	Curated list of major Indian MPAs	Manually compiled for the prototype
Cyclone, lightning	IMD public advisories	Not yet verified, needs a direct check

Important: avoid ISRO's own OCM-2 chlorophyll sensor for this build, it is marked "Restricted, as per Department of Space guidelines" in INCOIS's data holdings table, which likely means a formal approval process. Use MODIS or VIIRS instead, both confirmed open and real-time.

Action item before the hackathon: register on MOSDAC today, since full data access needs signup and email confirmation, which can take time.

7. System architecture — agents

Agent	Role
Planning Agent	Decomposes the user query into sub-tasks
Marine Data Discovery Agent	Finds and retrieves the right dataset for the query
Weather Intelligence Agent	Cyclone, lightning, wind, wave conditions, proactive alerts

Ocean Analytics Agent	Chlorophyll, SST, bloom detection, forecasting
Geospatial Reasoning Agent	Location resolution, EEZ and MPA geofencing, route safety
Risk Assessment Agent	Synthesises signals into a verdict with a confidence score
Visualization Agent	Chooses map, chart, heatmap, or advisory format for the answer
Reporting / User Interaction Agent	Conversational front-end, language detection, multi-turn memory

8. Tech stack

Layer	Technology	Why
Frontend	React + Tailwind CSS	Most widely known stack, fastest to debug under time pressure
Map	Leaflet	Free, no API key, standard for scientific and GIS tools, matches the Zoom Earth reference feel
Charts	Recharts	Simple React-native line charts with confidence bands
Backend	Python + FastAPI	Async-friendly, pairs directly with the Python ML and geospatial stack
Agent orchestration	Plain Python functions, no framework	Avoids losing hackathon time learning a new framework
LLM	Claude API	Handles reasoning, explanation, and multilingual response in one dependency
Forecasting	pandas / numpy trend extrapolation, LSTM if time allows	Start simple and working, improve only if time permits
Geospatial processing	xarray, rasterio, geopandas, shapely	Standard tools for satellite and geospatial data in Python

9. Stakeholders and real-world benefit

Stakeholder	What they see	Why it matters to them
Fishermen	Favourability verdict, bloom risk, forecast, route safety	Avoids wasted trips and fuel, avoids contaminated catch, points toward better fishing grounds
Coastal patrol / Coast Guard	Geofencing alerts, hazard-aware route planning	Better patrol planning and boundary awareness, as a complementary layer, not a surveillance system
Person on a sea trip / maritime operator	Combined safety verdict, route hazard check	One trustworthy answer instead of checking five separate sources
Local coastal communities	Plain-language hazard alerts in their own language	Early awareness of cyclones, high waves, and lightning risk
Researchers / coastal authorities	Historical trend and bloom pattern analysis	Faster starting point for investigating ecosystem changes

10. Requirement coverage against the official brief

Official requirement	Owning agent(s)
Understand natural language intent	Planning Agent
Auto-detect language, respond in kind	Reporting Agent
Multi-turn contextual conversation	Reporting Agent (session memory)
Discover and integrate multi-source data	Marine Data Discovery Agent
Spatial, temporal, contextual reasoning	Ocean Analytics + Geospatial Reasoning Agents
Explainable output via maps, charts, advisories	Visualization + Risk Assessment Agents

Proactive hazard alerts	Weather Intelligence Agent
Geofencing near boundaries and protected areas	Geospatial Reasoning Agent
Route optimisation and safe navigation	Geospatial + Weather Intelligence Agents
Reliable, evidenced recommendations	Risk Assessment Agent

11. What is still open — action items before the hackathon

- Assign team roles: frontend (map and chat UI), backend (FastAPI and agents), data pipeline (ISRO / INCOIS downloads).
- Register on MOSDAC today, do not wait until hackathon day.
- Confirm IMD cyclone and lightning data access.
- Prepare a demo script: 2 to 3 specific live questions to ask, plus a recorded backup video in case live data or API calls fail during judging.
- Validate the bloom detection logic against a known historical algal bloom event on India's coast, to show the model would have caught a real event.
- Prepare the slide-formatted pitch deck, this document is the content backbone for it.

12. Scope discipline — what we are intentionally not building

- Long-haul, multi-day voyage route planning (for example Kerala to West Bengal): mentioned as future scope only, since forecast confidence weakens over longer time horizons.
- Fish species identification: chlorophyll data indicates general productivity, not species presence. Only pursued as a stretch goal if historical catch-by-species data becomes available.
- Vessel tracking or surveillance capability for defence use: this is an environmental and geofencing awareness layer, not a tracking system, and should never be presented as one.